

Causal Prefix–Sinkhorn Attention: Leak-Free Column Balancing for Autoregressive Transformers

David Tom Foss  

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Abstract. Doubly stochastic attention—row- and column-normalized attention matrices obtained by Sinkhorn iteration—has shown optimization and rank-preservation benefits in bidirectional Transformers, but carrying the same column-balancing principle into the causal token-attention matrix is nontrivial. We first show that the naïve adaptation, running Sinkhorn iterations over the causally masked support, is *not causal*: the column-normalization step routes information from future rows into past rows. In a byte-level language-modeling A/B this defect is invisible in the loss curve and manifests instead as an impossibly good result (0.156 bits/byte), which a direct gradient probe exposes as future leakage (future-input gradient norm 0.385). We then propose *causal prefix–Sinkhorn attention*, which replaces global column sums with *prefix* column sums computed by a cumulative sum over rows: row i discounts each key by the attention mass that rows $i' \leq i$ have already allocated to it, followed by row renormalization. The construction is strictly causal—the gradient of any output with respect to any future input is exactly zero, which we verify to machine precision—and adds only a cumulative sum per iteration. Under an identical small-scale training budget it improves byte-level validation loss over softmax attention by 0.936 bits/byte (2.348 vs. 3.284 at step 400), and it trains stably over a 100× longer horizon. We position the operator as an *online* approximation of doubly stochastic balancing, relate it to attention-sink mitigation and rank-collapse theory, and describe the ablation campaign currently in progress.

1 Introduction

Softmax attention normalizes each *row* of the attention matrix independently. Nothing constrains the *columns*: a few keys may absorb attention mass from almost every query, a behavior closely related to the attention-sink phenomenon observed in language models [Gu et al., 2024, Ran-Milo, 2026], and nothing pulls the matrix toward the balanced, doubly stochastic class. That class is not an arbitrary target. In the author’s work on network algebra, double stochasticity is exactly the signature of *lossless* power transfer: the squared-magnitude matrix of any lossless scattering matrix is doubly stochastic, with a rich algebraic structure [Foss, 2026b,a], and random doubly stochastic matrices carry universal spectral statistics [Foss, 2026c,d]. Read through that lens, an attention matrix with balanced columns is one in which no key silently dissipates or hoards probability mass—a lossless router.

Lineage. This paper did not begin from the attention literature. It grew out of two lines of the author’s program: the algebra and spectral statistics of doubly stochastic matrices cited above, and prefix-scan state-space sequence models—GSSM’s bounded recurrence reduced to a cumulative sum [Foss, 2026e], its kernel unification of the linear state-space family [Foss, 2026g], and the constant-memory streaming philosophy of o1-state [Foss, 2026f]. Prior internal word-level experiments in this program measured large gains from Sinkhorn-style balancing of attention (up to 66% perplexity reduction at convergence, with a single balancing iteration carrying 86% of the effect). The construction below is the leak-free reconstruction of that finding, and it is the natural meeting point of the two lines: doubly stochastic balance, enforced as a prefix scan. A post-hoc novelty sweep (§5) located the nearest neighbors in the attention literature; none contributed to the construction. Balancing itself is classical [Sinkhorn, 1964]; in Transformers, Sinkformers [Sander et al., 2022] establish it for bidirectional attention, and Sparse Sinkhorn Attention [Tay et al., 2020] applies masked balancing to a *block permutation* matrix used for sorting—not to the token attention matrix itself.

This gap is not an accident. Column normalization couples every row of the matrix, including rows that correspond to *future* queries; making Sinkhorn-style balancing compatible with causal decoding therefore requires more than applying the causal mask inside the iteration. Our contributions:

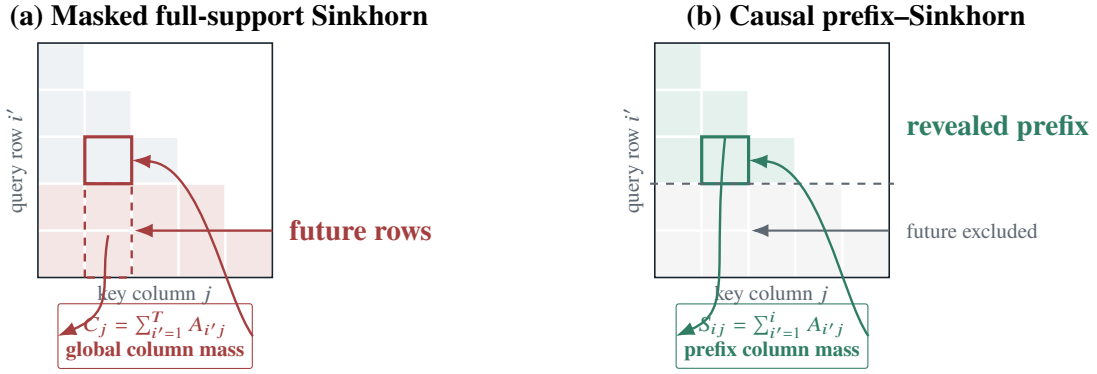


Figure 1. Why a triangular mask is insufficient. For a past query row $i = 3$ and key column $j = 2$, full-support column normalization (left) still aggregates entries from future query rows and feeds that denominator back into A_{ij} . Prefix–Sinkhorn (right) closes the column mass at the current row, so the normalizer is a function of the revealed prefix only. The schematic shows dependency flow, not numerical attention values.

1. **A falsification.** We show empirically that masked full-support Sinkhorn iterations leak future information in a decoder: the loss becomes impossibly good while a direct gradient probe shows a future-input gradient norm of 0.385 (§2). We advocate the probe itself as a cheap standard test for any nonstandard attention normalizer.
2. **A construction.** *Causal prefix–Sinkhorn attention* (§3): column masses are replaced by prefix column masses via a cumulative sum over rows, alternated with row renormalization. We prove strict causality and verify exact-zero future gradients numerically.
3. **Evidence.** Under an identical training budget on byte-level TinyStories, the operator improves held-out loss by 0.936 bpb over softmax attention, and trains stably for 76M bytes without collapse (§4), consistent with the program’s prior word-level measurements.
4. **Ongoing work.** A preregistered ablation campaign (scale, iteration count, column-norm-only, word level, attention-sink control) is running, with its falsifier—the gain vanishing with scale—stated explicitly (§6).

2 The leak in masked Sinkhorn

Setup. Let $\mathbf{L} \in \mathbb{R}^{T \times T}$ be the causal attention logits and $\mathbf{A}^{(0)} = \text{softmax}(\mathbf{L} + \mathbf{M})$ the row-softmax over the masked logits, with $\mathbf{M}_{ij} = 0$ for $j \leq i$ and $-\infty$ otherwise. Row i of $\mathbf{A}^{(0)}$ depends only on q_i and $k_{\leq i}$; softmax attention is causal.

Masked full-support balancing. The naïve causal Sinkhorn alternates, for $t = 1, \dots, k$,

$$\tilde{\mathbf{A}}_{ij} \leftarrow \frac{\mathbf{A}_{ij}}{\sum_{i'=1}^T \mathbf{A}_{i'j}}, \quad \mathbf{A}_{ij} \leftarrow \frac{[[j \leq i]] \tilde{\mathbf{A}}_{ij}}{\sum_{j'=[j \leq i]} \tilde{\mathbf{A}}_{ij'}}, \quad (1)$$

i.e. column normalization over the full (masked) support followed by row renormalization. The column sum $\sum_{i'} \mathbf{A}_{i'j}$ includes rows $i' > i$: the normalized entry at row i depends on queries of *future* positions. The mask makes the *matrix* lower triangular, but not the *computation* causal.

Empirical falsification. We trained two identical byte-level Transformer LMs (§4 for the configuration), one with softmax attention and one with (1). The leaky model reached 0.156 bpb held-out at small budget—far below any plausible value for the model class, and “too good to be true” was the correct reading: a direct probe (backpropagating the loss at position i to the inputs at positions $j > i$) gives a future-input gradient norm of 0.385 for (1), versus exactly 0 for softmax. The model was not learning language; it was reading the future through the normalizer. We discarded the run and archived it as a negative result.

The probe. The test is three lines: forward on a random batch, loss at a single position i , backward, then inspect $\|\partial \mathcal{L}_i / \partial x_{j>i}\|$. We require *exact* zero (below 10^{-12} in float32) for any normalizer we train with, and suggest this as a standard acceptance test for nonstandard attention normalizations. Notably, the closest prior construction—“causal Sinkhorn balancing” for sorting networks [Tay et al., 2020, §3.3.2]—normalizes over the masked support in the style of (1); whether that use case (permutation logits over blocks) leaks is outside our scope, but for token-level attention the pattern fails the probe.

3 Causal prefix–Sinkhorn attention

Construction. Replace the global column mass in (1) by the *prefix* column mass, computable with a cumulative sum over rows. Starting from the row-softmax $\mathbf{A}^{(0)}$, iterate for $t = 1, \dots, k$:

$$\begin{aligned} S_{ij} &= \sum_{i'=1}^i \mathbf{A}_{i'j} \quad (\text{cumulative sum over rows}), \\ \mathbf{A}_{ij} &\leftarrow \frac{\mathbf{A}_{ij}}{S_{ij} + \varepsilon}, \quad \mathbf{A}_{ij} \leftarrow \llbracket j \leq i \rrbracket \mathbf{A}_{ij}, \quad \mathbf{A}_{ij} \leftarrow \frac{\mathbf{A}_{ij}}{\sum_{j'} \mathbf{A}_{ij'} + \varepsilon}. \end{aligned} \quad (2)$$

The output is $\mathbf{AV}$ as usual. One iteration adds a cumulative sum, two elementwise operations, and a row normalization; the asymptotic cost remains $O(T^2)$ per head and the cumulative sum is a parallel scan.

Proposition 1 (Strict causality). *For every iteration count k , row i of the matrix produced by (2) is a function of rows $1..i$ of $\mathbf{A}^{(0)}$ only. Consequently the layer output at position i depends only on tokens $1..i$, and $\partial(\text{output}_i) / \partial(\text{input}_j) = 0$ for all $j > i$.*

Proof. By induction over operations. $\mathbf{A}^{(0)}$ is causal (row softmax over masked logits). The prefix sum S_{ij} involves rows $i' \leq i$ only; elementwise division leaves row i a function of rows $\leq i$; the mask is constant; row normalization uses row i only. Composition over k iterations preserves the property, and $\mathbf{AV}$ reads values $v_{j \leq i}$ through the lower-triangular $\mathbf{A}$. $\square$

We verify Proposition 1 numerically with the probe of §2: the future-input gradient is 0.0 to machine precision (float32) for all tested k , versus 0.385 for (1) (Table 2).

Interpretation: online balancing. S_{ij} is the attention mass that the *prefix* $i' \leq i$ has already spent on key j . Dividing by S_{ij} makes row i discount keys in proportion to their past usage—an implicit per-key budget enforced in decoding order, followed by row renormalization so each query still distributes one unit of mass. Full doubly stochastic balancing enforces unit column sums using the whole sequence; (2) enforces the analogous pressure using only the revealed prefix, making it an *online* (anytime) approximation of the Sinkhorn projection rather than a masked corruption of it. In the DS^2 picture [Foss, 2026b], unit column sums are precisely the balance property of lossless power matrices: the operator pulls attention toward the lossless-router class using only what decoding has revealed, targeting the two familiar pathologies of softmax attention—keys that hoard column mass, and the low-rank drift of deep attention stacks [Dong et al., 2021]. Early rows see small prefixes ($S_{1j} = \mathbf{A}_{1j}$, so row 1 is renormalized to itself), so balancing pressure grows with position; the interaction of this schedule with learned behavior is measured in the ablations (§6).

4 Experiments

Configuration. Byte-level language modeling (vocabulary 256) on TinyStories [Eldan and Li, 2023]; decoder-only Transformer with 4 blocks, width $d=192$, 4 heads, context 256, learned positional embeddings; AdamW, learning rate $3 \cdot 10^{-4}$, weight decay 0.01, gradient clip 1.0, batch 24; identical seeds and data order across arms. The balancing uses $k=1$ iteration (all numbers below; prior internal word-level experiments found one iteration carries most of the effect). CPU training (the run is deliberately reproducible on commodity hardware).

Attention	Causal?	Held-out bpb @ 400	Δ vs. softmax
Softmax (row-normalized)	yes	3.284	—
Masked Sinkhorn, Eq. (1)	no	0.156 [†]	invalid
Prefix–Sinkhorn, Eq. (2), $k=1$	yes	2.348	−0.936

Table 1. Identical-budget A/B on byte-level TinyStories. [†]The masked-Sinkhorn value is deliberately retained as a falsification signal: its implausibly low loss is produced by future leakage and is not a valid language-modeling result. Lower is better.

Attention	Future-input gradient norm
Softmax	0.0 (exact)
Masked Sinkhorn, Eq. (1)	0.385
Prefix–Sinkhorn, Eq. (2)	0.0 (exact; $< 10^{-12}$)

Table 2. Direct leak probe: backpropagate a loss at position i and inspect gradients with respect to inputs at $j > i$. The probe cleanly separates triangular support from computational causality.

Longer horizon. The same $k=1$ configuration has been training continuously on a CPU server: held-out 1.100 bpb after 76M bytes (step 12,400), with smooth loss curves and no instability attributable to the normalizer. For calibration, a tuned recurrent byte-level baseline from the same lab and data reaches 1.088 bpb only after 95M bytes; per byte seen, the prefix–Sinkhorn Transformer is currently ahead while streaming 2.3× more bytes per second in our implementation.

5 Positioning and independence

Research lineage. The construction was developed within the author’s program on doubly stochastic power matrices and their algebraic/spectral structure [Foss, 2026b,a,c,d], together with prefix-scan state-space models [Foss, 2026e,g,f]. The operational move in Eq. (2)—replacing a global reduction by a cumulative prefix reduction—comes directly from that scan-based line. The empirical seed was the program’s earlier word-level Sinkhorn-attention measurements.

Doubly stochastic attention. Sinkformers [Sander et al., 2022] introduced Sinkhorn-normalized, doubly stochastic token attention and demonstrated it in bidirectional settings. More recent work has expanded this design space: ESPFormer replaces iterative scaling with expected sliced transport plans [Shahbazi et al., 2025]; LOTFormer constructs a low-rank transport factorization with linear-time application [Shahbazi et al., 2026]; and ASAP amortizes a trained Sinkhorn normalizer for cheaper repeated inference [Tran et al., 2026]. A 2026 analysis by Lapenna et al. [2026] directly compares rank decay under Softmax and Sinkhorn normalization and finds that doubly stochastic attention delays rank collapse empirically, while pure self-attention still approaches rank one with depth. These works strengthen the case for column balancing but are formulated around bidirectional or non-causal attention; notably, both LOTFormer and ASAP explicitly leave causal marginal design as a separate problem.

Causality and the closest naming collision. Sparse Sinkhorn Attention [Tay et al., 2020] proposes “causal Sinkhorn balancing” for autoregressive decoding, but the Sinkhorn matrix there is a *block-permutation/sorting* object that determines which blocks are brought into local attention. It is not the token-to-token attention matrix balanced in this paper. Applying full-support column normalization directly to a causally masked token-attention matrix yields Eq. (1) and fails the zero-future-gradient probe. Our novelty claim is therefore narrow: *prefix-mass column balancing of the causal token-attention matrix, with a zero-future-gradient guarantee*. A targeted literature sweep through August 2026 found no direct instance of that operator.

Attention sinks and rank collapse. The motivation is also adjacent to two well-studied failure modes of standard attention. Pure self-attention has a strong rank-one bias with depth [Dong et al., 2021].

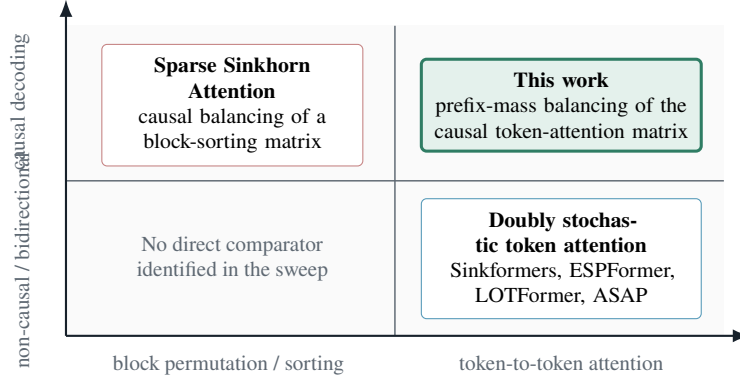


Figure 2. The relevant design space separates *what* is balanced from *where causality is enforced*. Sparse Sinkhorn Attention balances a causal block-sorting/permutation object [Tay et al., 2020]; Sinkformers and its recent descendants balance token attention but operate in bidirectional/non-causal settings [Sander et al., 2022, Shahbazi et al., 2025, 2026, Tran et al., 2026]. Prefix–Sinkhorn occupies the causal token-attention quadrant. The map is a taxonomy, not a performance ranking.

Attention sinks are empirically widespread and have been linked in part to Softmax normalization [Gu et al., 2024]; recent theory gives tasks for which a Softmax Transformer necessarily develops a sink while a non-normalized alternative need not [Ran-Milo, 2026]. Prefix–Sinkhorn targets a different object—column mass rather than an explicit sink token—but its per-key prefix budget provides a concrete mechanism by which persistent mass concentration can be discouraged.

6 Ongoing work

A preregistered ablation campaign is running: scale at $d \in \{128, 256, 384\}$ (two seeds, softmax and prefix–Sinkhorn pairs at $k=1$), an added-constant softmax denominator (letting heads abstain—tests whether the gain reduces to sink mitigation), column-norm-only (no row renormalization inside the loop), $k \in \{2, 3, 5\}$, and word-level tokenization. Per-run instrumentation records row entropies, column-mass Gini, effective rank, and the deviation of attention spectra from the universal baselines of Foss [2026d] and Foss [2026c] (“Ginibre- Δ ”), computed with the DS² tooling [Foss, 2026a]. The falsifier is stated explicitly: the gain vanishing with scale (Δ bpb $\rightarrow 0$ as d grows). A scan-form for $k=1$ KV-cache decoding (maintain per-key prefix masses) exists and will be reported with the ablations.

Reproducibility

The reference implementation (a ~160-line training script), the leak probe, all seeds, and the run logs live in the author’s private repository; the A/B of Table 1 trains in minutes on a laptop CPU. The draft will be revised once the ablation campaign completes.

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